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# Is a tensor with the SAME support and moduli as T but the TRIVIAL sign character
invariant under the 28-dim algebra?
import numpy as np, io, contextlib
with contextlib.redirect_stdout(io.StringIO()):
exec(open('39_robustness_of_T_sign_structure.py').read())
T=T0; cols=[]
for s in range(3):
    for M in Sb:
        r=[np.einsum('ai,ajk->ijk',M,T),np.einsum('bj,ibk-
>ijk',M,T),np.einsum('ck,ijc->ijk',M,T)][s]; cols.append(r.ravel())
u,sv,vt=np.linalg.svd(np.array(cols).T); null=vt[np.sum(sv>TOL):]; d=null.shape[0];
L=[trip(v) for v in null]
cm=lambda X,Y:[X[s]@Y[s]-Y[s]@X[s] for s in range(3)]
co=lambda t: null@np.concatenate([np.array([M[p,q] for p in range(8) for q in
range(p+1,8)]) for M in t])
ad=lambda X: np.array([co(cm(X,L[b])) for b in range(d)]).T
rng=np.random.RandomState(0); X=trip(rng.randn(d)@null);
_,s2,v2=np.linalg.svd(ad(X)); cent=v2[np.sum(s2>1e-9):]; Hs=[trip(c@null) for c in
cent]
co_=rng.randn(len(Hs)); V=[]
for s in range(3):
    Hg=sum(c*H[s] for c,H in zip(co_,Hs)); w,VV=np.linalg.eig(Hg);
V.append(VV/np.linalg.norm(VV,axis=0))
Tw=np.einsum('abc,ai,bj,ck->ijk',T,V[0],V[1],V[2]); idx=support(Tw); Vi=
[np.linalg.inv(v) for v in V]
back=lambda A: np.einsum('ijk,ia,jb,kc->abc',A,Vi[0],Vi[1],Vi[2])
print("reconstruction of T from its weight-basis array: max error
%.1e"%np.abs(back(Tw)-T).max())
Tt=np.zeros_like(Tw)
for e in idx: Tt[e]=np.sqrt(2)                    # same support, same moduli, all
phases removed (trivial character)
S=back(Tt)
def res(Y,A): return np.abs(np.einsum('ai,ajk->ijk',Y[0],A)+np.einsum('bj,ibk-
>ijk',Y[1],A)+np.einsum('ck,ijc->ijk',Y[2],A)).max()
print("invariance residual of T itself over the 28 algebra elements: max
%.1e"%max(res(Lx,T) for Lx in L))
print("invariance residual of the trivial-character tensor:          max %.2f   (norm
of tensor %.2f)"%(max(res(Lx,S) for Lx in L),np.linalg.norm(S)))
c=np.vdot(T.ravel(),S.ravel())/np.vdot(T.ravel(),T.ravel()); print("distance of the
trivial-character tensor from the line spanned by T (relative): %.3f"%
(np.linalg.norm(S-c*T)/np.linalg.norm(S)))
# sign-flipped single entry (changes the character) as a second comparison
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Tf=Tw.copy(); Tf[idx[0]]*=-1; print("T with ONE entry's sign flipped: invariance
residual max %.2f"%max(res(Lx,back(Tf)) for Lx in L))
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